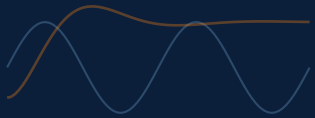
DAY 4 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



Servo Systems: Drives, FOC, Control Loops, Tuning and Compensation

3.2 PWM & limits ▪ 3.3 Field-oriented control ▪ 3.4 Cascade loops ▪ 3.5 Tuning & bandwidth ▪ 3.6 Disturbance compensation

Mechatronics Training Programme

Theoretical Course Series ▪ Day 4 of 5

Agenda and learning objectives

Code	Session	Net time
3.2	(cont.) PWM, modulation range, dead time, limits	0.5 h
3.3	Field-oriented control of the PMSM	1.5 h
3.4	Current, velocity and position loops; trajectories	1.5 h
3.5	Loop tuning and bandwidth	1.0 h
3.6	Disturbances and compensation	0.5 h
	Total (+ 10-min breaks about every hour)	**5.0 h**

Reference motor used all day (surface-mounted PMSM servo): $p = 4$, $R = 0.5\,\Omega$, $L_d = L_q = 3\,\text{mH}$, $\psi_{\text{PM}} = 0.12\,\text{Vs}$, $k_t = \frac{3}{2} p \psi_{\text{PM}} = 0.72\,\text{Nm/A}$, $J_m = 3 \times 10^{-4}\,\text{kg m}^2$, load inertia ratio 3, $V_{\text{dc}} = 325\,\text{V}$, $\hat{I}_{\text{max}} = 20\,\text{A}$, $f_{\text{PWM}} = f_s = 16\,\text{kHz}$.

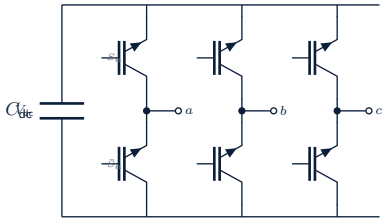
By the end of the day you can ...

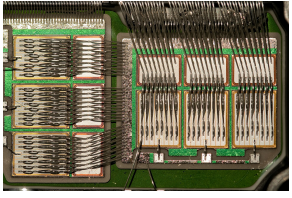
- compute SVPWM dwell times and the limit $V_{\text{dc}}/\sqrt{3}$
- transform $abc \to \alpha\beta \to dq$; write the PMSM dq model
- explain the FOC structure, MTPA and field weakening
- design PI current/velocity loops (TO / SO)
- size bandwidths; quantify the phase cost of delays
- generate motion profiles; cancel following error by feedforward
- explain disturbance observers and cogging compensation

Session 1

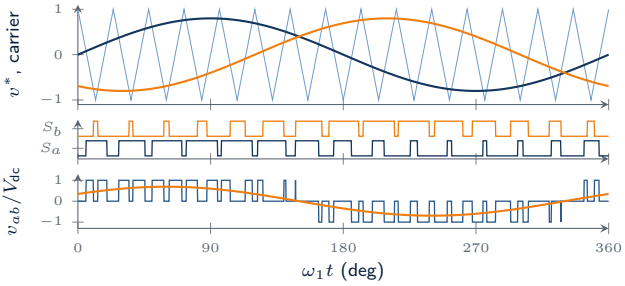
3.2 (cont.) Motor Driving: PWM, Voltage and Current Limits

The two-level voltage-source inverter and carrier-based sinusoidal PWM



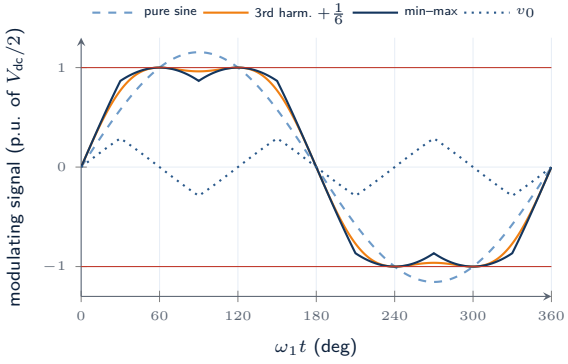


Pslawinski, CC BY-SA 3.0, via Wikimedia Commons



Compare $v_a^* = m \sin \omega_1 t$ with a triangle carrier ($m = 0.8$, $m_f = 15$): leg a ON when $v_a^* >$ carrier. Pole voltage average $\bar{v}_{aN} = \frac{V_{dc}}{2}(1 + v_a^*) \Rightarrow$ fundamental phase amplitude $\hat{V}_1 = m\, V_{dc}/2$.

Zero-sequence injection: third harmonic and min–max (SVPWM-equivalent)



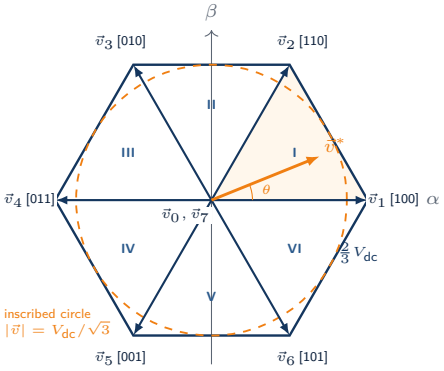
All curves for $\hat{V}_1 = \frac{2}{\sqrt{3}} \frac{V_{dc}}{2} = V_{dc}/\sqrt{3}$: the pure sine saturates, the injected signals just fit.

- A common-mode term v_0 added to all three references **cancels in the line-to-line voltages** (isolated star point).
- Third-harmonic injection: $v_x^* = m[\sin\theta_x + \frac{1}{6}\sin 3\theta]$
- Min–max injection: $v_0 = -\frac{1}{2}[\max(v^*) + \min(v^*)]$ – identical average voltages to centred SVPWM.

$\hat{V}_{1,\max}$: $V_{dc}/2$ (SPWM) $\rightarrow$ $V_{dc}/\sqrt{3}$ (THI, SVPWM)
$= +15.5\,\%$ linear voltage range

Beyond: **overmodulation** up to six-step ($\hat{V}_1 = \frac{2}{\pi} V_{dc}$), with low-order harmonics and non-linear gain.

Space vectors: the inverter hexagon



- Switching state $[S_a S_b S_c]$ gives the space vector $\vec{v} = \frac{2}{3}\left(v_{aN} + \underline{a}\, v_{bN} + \underline{a}^2 v_{cN}\right),\ \underline{a} = e^{j2\pi/3}$
- 6 active vectors of length $\frac{2}{3} V_{\text{dc}}$ and 2 zero vectors $(000,\ 111)$.
- 6 sectors of $60°$; adjacent vectors differ in **one leg only** – one commutation per transition.
- Any $\vec{v}^*$ inside the hexagon is synthesised *on average* over T_s from the two adjacent active vectors plus zero vectors.
- Rotating $\vec{v}^*$ without distortion: stay inside the inscribed circle, radius $\frac{2}{3} V_{\text{dc}} \cos 30° = V_{\text{dc}}/\sqrt{3}$.

[van der Broeck, Skudelny & Stanke, 1988]; [Holmes & Lipo, 2003]

SVPWM dwell times and the centred seven-segment sequence

Volt-second balance in sector I:

$$\vec{v}^{*} T_s = \vec{v}_1 T_1 + \vec{v}_2 T_2 + \vec{v}_{0/7} T_0$$

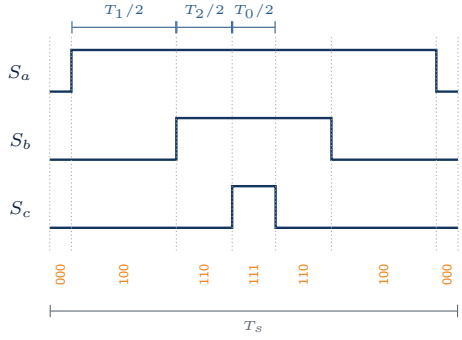
$$T_1 = \sqrt{3}\,\frac{|\vec{v}^{*}|}{V_{\mathrm{dc}}}\,T_s \sin(60^\circ - \theta)$$

$$T_2 = \sqrt{3}\,\frac{|\vec{v}^{*}|}{V_{\mathrm{dc}}}\,T_s \sin\theta$$

$$T_0 = T_s - T_1 - T_2 \geq 0$$

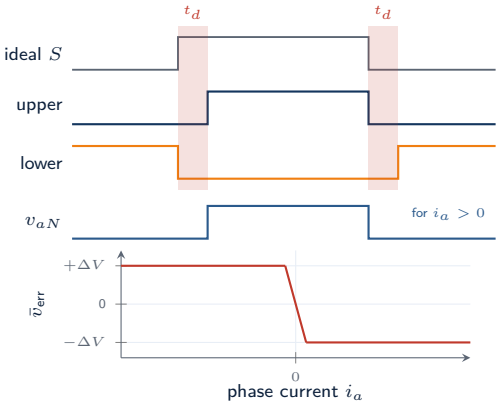
Worked example

$V_{\mathrm{dc}} = 325\,\mathrm{V}$, $|\vec{v}^{*}| = 150\,\mathrm{V}$ ($m = |\vec{v}^{*}|\sqrt{3}/V_{\mathrm{dc}} = 0.80$), $\theta = 20^\circ$, $T_s = 62.5\,\mu\mathrm{s}$:
$T_1 = 0.800 \cdot 0.643\,T_s = 32.1\,\mu\mathrm{s}$, $T_2 = 0.800 \cdot 0.342\,T_s = 17.1\,\mu\mathrm{s}$,
$T_0 = 13.3\,\mu\mathrm{s}$ split equally between 000 and 111.



Centred (symmetric) pattern, drawn to scale for the example: one switching per leg per half-period, samples taken in the middle of $000/111$ see the **average** current (ripple-free sampling).

Dead time: why it is needed and what it costs



- Both switches of a leg must never conduct together $\Rightarrow$ the turn-on edge is delayed by t_d (≈ 0.5–3 µs for IGBTs).
- During t_d the diodes conduct: the pole voltage depends on the **sign of the current**, not on the gate command.

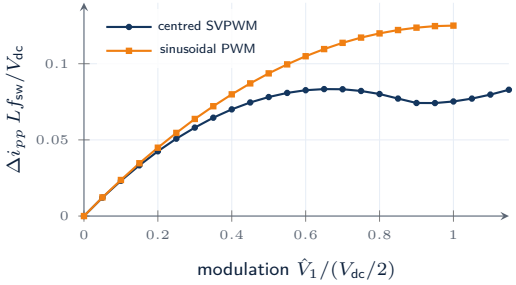
$$\Delta \bar{V} \approx t_d \, f_{\text{sw}} \, V_{\text{dc}} \;\; \text{sign}(i)$$

Example

$t_d = 1\,\mu\text{s}$, $f_{\text{sw}} = 16\,\text{kHz}$, $V_{\text{dc}} = 325\,\text{V}$: $\Delta \bar{V} = 5.2\,\text{V}$. At 100 rpm the motor needs only $\approx 5\,\text{V}$ back-EMF: the error is of the *same order* as the command!

- Symptoms: current clamping at zero crossings, 6th-harmonic ripple.
- Cure: dead-time compensation (add $\Delta V \, \text{sign}(i^*)$, smoothed near zero) and the current loop itself.

Switching frequency trade-offs and current ripple



Simulated worst-case peak-to-peak phase-current ripple over a fundamental period (ideal switching). Example: $L = 3\,\text{mH}$, $V_{dc} = 325\,\text{V}$, $f_{sw} = 16\,\text{kHz}$, ratio $0.083 \Rightarrow \Delta i_{pp} \approx 0.56\,\text{A}$.

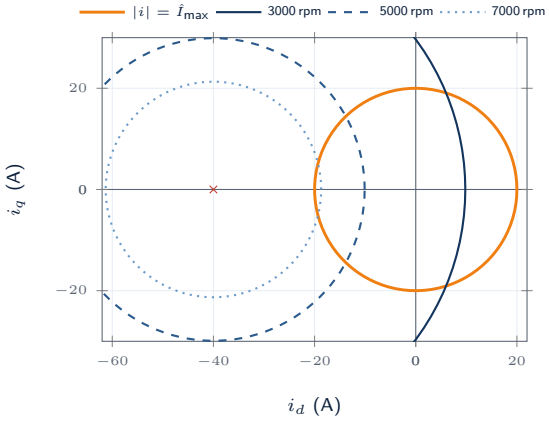
	higher f_{sw}	**lower f_{sw}**
ripple $\propto 1/f_{sw}$	smaller	larger
switching loss $\propto f_{sw}$	higher	lower
loop delay $\propto 1/f_s$	shorter	longer
current bandwidth	higher	lower
EMI, dv/dt events	more	fewer
acoustic noise	inaudible $> 16\,\text{kHz}$	audible

$$P_{sw} \approx 6 \cdot \tfrac{1}{\pi}\, f_{sw}\, E_{sw,ref} \tfrac{V_{dc}}{V_{ref}} \tfrac{\hat{I}}{I_{ref}}$$

Typical servo drives: **8–16 kHz** (IGBT, kW range); SiC/GaN allow 40–100 kHz with low loss.

[Mohan, Undeland & Robbins, 2003]

Voltage and current limits in the dq plane



Current limit: $i_d^2 + i_q^2 \le \hat{I}_{\max}^2$

Voltage limit (steady state, $R \approx 0$):

$$(L_d i_d + \psi_{\mathrm{PM}})^2 + (L_q i_q)^2 \le (V_{\max}/\omega_e)^2$$

$V_{\max} = V_{\mathrm{dc}}/\sqrt{3}$

- SPM ($L_d = L_q$): voltage circles centred at $-\psi_{\mathrm{PM}}/L_d = -40$ A (red cross), radius shrinks $\propto 1/\omega_e$.
- Feasible operating points: **intersection** of both regions.
- At 3000 rpm the whole current circle is reachable; at 5000 rpm i_q is only possible with negative i_d (field weakening); at 7000 rpm almost nothing is left.
- Keep a voltage reserve of 5–10 % for the current controller transients.

[Morimoto et al., 1990]

Summary 3.2 – PWM and limits

Key take-aways

- SPWM: $\hat{V}_1 \leq V_{\mathrm{dc}}/2$; zero-sequence (THI, min–max, SVPWM): $\hat{V}_1 \leq V_{\mathrm{dc}}/\sqrt{3}$.
- SVPWM = volt-second balance with two adjacent active vectors + zero vectors.
- Dead time creates a current-sign-dependent voltage error $t_d f_{\mathrm{sw}} V_{\mathrm{dc}}$ – dominant at low speed.
- f_{sw}: ripple and delay vs. switching loss and EMI.
- Every operating point must satisfy both the current circle and the speed-dependent voltage limit.

Self-check

1. $V_{\mathrm{dc}} = 560\,\mathrm{V}$: maximum linear phase amplitude? (323 V)
2. Why does the zero-sequence term not appear in the motor currents?
3. Which vectors are used in sector IV? ($\vec{v}_4, \vec{v}_5$)

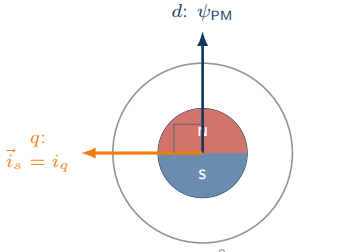
Session 2

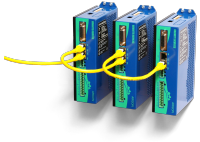
3.3 Field-Oriented Control (FOC)

Why field-oriented control? The DC-machine analogy

- Brushed DC motor: commutator keeps armature MMF **orthogonal** to the field $\Rightarrow T = k_t\, i_a$, linear, instantaneous.
- PMSM: the rotor flux rotates; the stator current is three AC quantities.
- FOC = **electronic commutator**: measure θ_e, express the current in a frame fixed to the rotor flux, and control
  - i_d – aligned with ψ_{PM} (flux-producing),
  - i_q – 90° ahead (torque-producing).
- In the rotor frame all sinusoidal quantities become **DC** in steady state $\Rightarrow$ PI controllers reach zero steady-state error at any speed.
- Requires: fast current sensing, accurate rotor angle, and a modulator (SVPWM).

[Blaschke, 1972]; [Leonhard, 2001]; [Krishnan, 2010]





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Clarke and Park transforms (amplitude-invariant form)

Clarke: $abc \to \alpha\beta$ (stationary)

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

With $i_a + i_b + i_c = 0$: $i_\alpha = i_a,\ i_\beta = (i_a + 2i_b)/\sqrt{3}$ – only two sensors needed.

Park: $\alpha\beta \to dq$ (rotating with θ_e)

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos\theta_e & \sin\theta_e \\ -\sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}$$

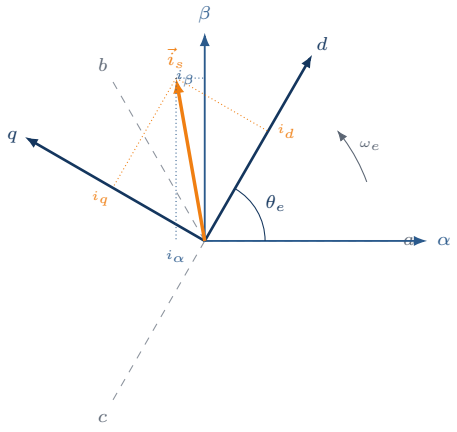
- Factor $\frac{2}{3}$: **amplitude-invariant** – a balanced set of peak $\hat{I}$ gives $|\vec{i}| = \hat{I}$.
- Price: power is not invariant:

$$p = \tfrac{3}{2}\left(v_\alpha i_\alpha + v_\beta i_\beta\right) = \tfrac{3}{2}\left(v_d i_d + v_q i_q\right)$$

- Power-invariant variant uses $\sqrt{2/3}$ instead of $\frac{2}{3}$ (no $\frac{3}{2}$ in p and T) – check which one a drive manual uses!
- Inverse Park ($dq \to \alpha\beta$) uses the transposed rotation; inverse Clarke: $i_a = i_\alpha$, $i_{b,c} = -\frac{1}{2} i_\alpha \pm \frac{\sqrt{3}}{2} i_\beta$.
- Zero-sequence $i_0 = \frac{1}{3}(i_a + i_b + i_c) = 0$ for a star without neutral.

[Park, 1929]; [Clarke, 1943]

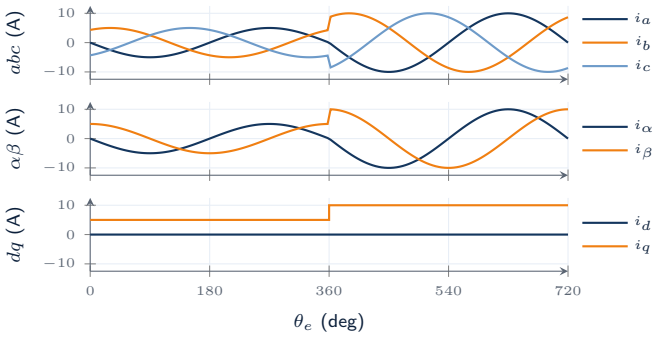
Reference frames: abc, $\alpha\beta$ and dq



- abc: physical phase axes, $120°$ apart (windings).
- $\alpha\beta$: orthogonal, stationary; $\alpha \equiv a$. Quantities rotate at ω_e.
- dq: orthogonal, rotates with the rotor, d on the magnet north pole: $\theta_e = \int \omega_e \, dt$.
- The current *vector* is the same physical object in all three frames – only the coordinates change.
- Space-vector notation:

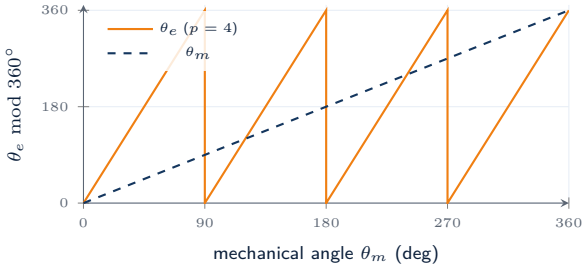
$$\vec{i}_s^{\,dq} = \vec{i}_s^{\,\alpha\beta}\, e^{-j\theta_e}, \qquad \vec{i}_s^{\,\alpha\beta} = \tfrac{2}{3}\left(i_a + \underline{a}\, i_b + \underline{a}^2 i_c\right)$$

The same current seen in the three frames



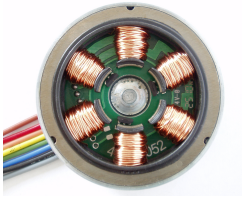
- Balanced currents $i_a = \hat{I}\cos(\theta_e + 90^\circ)$ etc. with a step $\hat{I} : 5 \to 10\,\text{A}$ at $\theta_e = 360^\circ$.
- $\alpha\beta$: still sinusoidal, frequency ω_e.
- dq: **DC**: $i_d = 0$, $i_q = \hat{I}$ – the step is a clean step.
- A PI controller in dq therefore tracks a sinusoid without phase lag at any speed (it is an *integrator at* ω_e in $\alpha\beta$).

Electrical vs. mechanical angle: $\theta_e = p\,\theta_m$



$$\theta_e = p\,\theta_m, \quad \omega_e = p\,\omega_m, \quad f_e = \frac{p\,n}{60}$$

Example: $p = 4$, $n = 3000$ rpm $\Rightarrow$ $f_e = 200$ Hz, $\omega_e = 1257$ rad/s.



12-slot stator. Medvedev, CC BY-SA 3.0, via Wikimedia Commons

- p = number of *pole pairs* (an 8-pole motor has $p = 4$).
- Encoder resolution in electrical degrees is p times coarser.
- A mechanical alignment error $\Delta\theta_m$ becomes $p\,\Delta\theta_m$ in the Park transform.

PMSM model in rotor (dq) coordinates

Stator equation in stationary coordinates (space vectors):

$$\vec{v}_s = R\,\vec{i}_s + \frac{d\vec{\psi}_s}{dt}$$

Substitute $\vec{x} = \vec{x}^{\,dq} e^{j\theta_e}$:

$$\vec{v}^{\,dq} = R\,\vec{i}^{\,dq} + \frac{d\vec{\psi}^{\,dq}}{dt} + j\omega_e \vec{\psi}^{\,dq}$$

Flux linkages (rotor frame, no saturation):

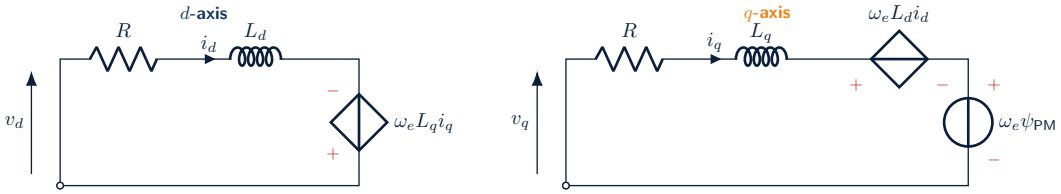
$$\psi_d = L_d i_d + \psi_{\mathrm{PM}}, \qquad \psi_q = L_q i_q$$

$$v_d = R\,i_d + L_d \frac{di_d}{dt} \underbrace{-\,\omega_e L_q i_q}_{\text{cross-coupling}}$$

$$v_q = R\,i_q + L_q \frac{di_q}{dt} \underbrace{+\,\omega_e L_d i_d}_{\text{cross-coupling}} \underbrace{+\,\omega_e \psi_{\mathrm{PM}}}_{\text{back-EMF}}$$

- Units: V, A, H, Vs; ω_e in rad/s (electrical). Motor sign convention (power into terminals positive).
- The rotational term $j\omega_e\vec{\psi}$ produces the **speed-dependent coupling** between axes and the back-EMF on the q-axis only.
- SPM: $L_d \approx L_q = L$; IPM: $L_q > L_d$ (saliency).
- Mechanics: $J\dot{\omega}_m = T_e - B\omega_m - T_L$.

Equivalent circuits of the d- and q-axis



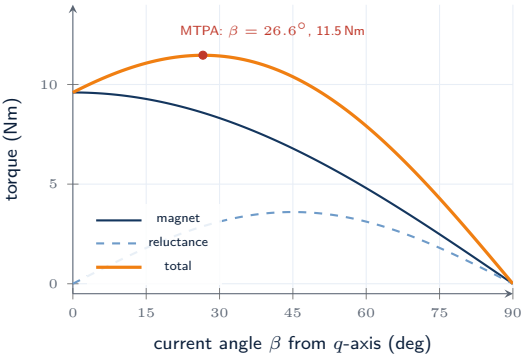
red signs: polarity of each speed voltage as a voltage drop in the direction of the current

- Each axis is an R–L **circuit** with a speed-dependent voltage source.
- Electrical time constants $\tau_d = L_d/R$, $\tau_q = L_q/R$ (here 6 ms).
- The sources are **disturbances** for the current loops: known (measured i, ω_e) $\Rightarrow$ can be fed forward (decoupling).
- q-axis carries the back-EMF; it sets the voltage requirement at speed.

Torque: flux- and torque-producing currents, reluctance torque

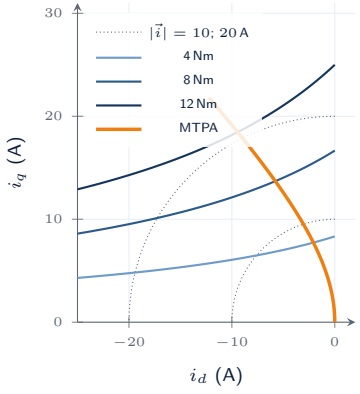
$$T_e = \tfrac{3}{2}\, p\, (\psi_d i_q - \psi_q i_d)$$
$$= \tfrac{3}{2}\, p \left[\underbrace{\psi_{\mathrm{PM}}\, i_q}_{\text{magnet}} + \underbrace{(L_d - L_q)\, i_d i_q}_{\text{reluctance}} \right]$$

- SPM ($L_d = L_q$): $T_e = k_t i_q$, $k_t = \frac{3}{2} p \psi_{\mathrm{PM}}$ (0.72 Nm/A for the reference motor); $i_d^* = 0$ minimises $|\vec{i}|$.
- IPM ($L_q > L_d$): **negative i_d adds torque**.
- With $i_d = -\hat{I} \sin\beta$, $i_q = \hat{I} \cos\beta$: $T_e = \frac{3}{2} p \left[\psi_{\mathrm{PM}} \hat{I} \cos\beta + \frac{1}{2} (L_q - L_d) \hat{I}^2 \sin 2\beta \right]$
- Note: k_t in Nm/A refers to the *peak* (amplitude-invariant) current; datasheets often use rms: $k_{t,\mathrm{rms}} = \sqrt{2}\, k_t$.



Illustrative IPM: $p = 4$, $\psi_{\mathrm{PM}} = 0.08$ Vs, $L_d = 2$ mH, $L_q = 5$ mH, $\hat{I} = 20$ A (+19 % vs. $i_d = 0$).

Maximum torque per ampere (MTPA)



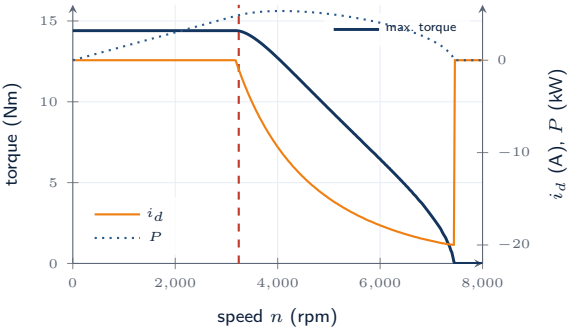
- Goal: minimum $|\vec{i}|$ (copper loss, inverter current) for a required torque.
- Condition: constant-torque curve **tangent** to the current circle. Setting $\partial T_e / \partial \beta = 0$:

$$i_d^{\mathrm{MTPA}} = \frac{\psi_{\mathrm{PM}}}{4(L_q - L_d)} - \sqrt{\frac{\psi_{\mathrm{PM}}^2}{16(L_q - L_d)^2} + \frac{\hat{I}^2}{2}}$$

- SPM: $L_q = L_d \Rightarrow$ MTPA is simply $i_d = 0$.
- Implementation: look-up table $i_d^* = f(i_q^*)$ or $(i_d^*, i_q^*) = f(T^*)$; tables measured on a dynamometer include saturation (L_q drops with i_q).

[Jahns, Kliman & Neumann, 1986]

Field weakening (brief) and the torque–speed envelope



Computed for the reference SPM ($V_{\text{dc}} = 325\,\text{V}$, $\hat{I}_{\text{max}} = 20\,\text{A}$): base speed ≈ 3240 rpm, limit ≈ 7440 rpm.

- Above base speed the back-EMF $\omega_e \psi_{\text{PM}}$ approaches V_{max}.
- Negative i_d **reduces the d-axis flux** $L_d i_d + \psi_{\text{PM}}$ $\Rightarrow$ more speed at the same voltage.
- The current budget used by i_d is lost for i_q: torque falls, power roughly constant.
- Characteristic current $I_{\text{ch}} = \psi_{\text{PM}}/L_d$ (here $40\,\text{A} > \hat{I}_{\text{max}}$ $\Rightarrow$ finite maximum speed).
- Practical: voltage-feedback FW loop ($|\vec{v}^{*}| > 0.95 V_{\text{max}}$ $\Rightarrow$ integrate towards negative i_d^{*}).
- Servo motors seldom use deep FW; spindles and traction do.

Worked example: steady-state operating point at 3000 rpm

Reference SPM, $i_d = 0$, $i_q = 10\,\text{A}$, $n = 3000\,\text{rpm}$:

$$\omega_e = p\,\frac{2\pi n}{60} = 4 \cdot 314.2 = 1256.6\,\text{rad/s}$$

$$E = \omega_e \psi_{\text{PM}} = 150.8\,\text{V}$$

$$v_q = R\,i_q + E = 5.0 + 150.8 = 155.8\,\text{V}$$

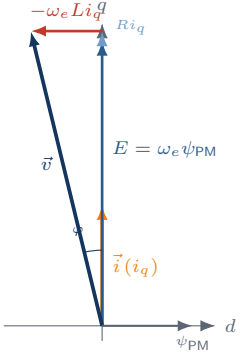
$$v_d = -\omega_e L\,i_q = -37.7\,\text{V}$$

$$|\vec{v}| = \sqrt{v_d^2 + v_q^2} = 160.3\,\text{V}\ (85\,\%\ \text{of}\ V_{\text{max}} = 187.6\,\text{V})$$

$$T_e = k_t i_q = 7.2\,\text{Nm}, \quad P_{\text{mech}} = T_e \omega_m = 2.26\,\text{kW}$$

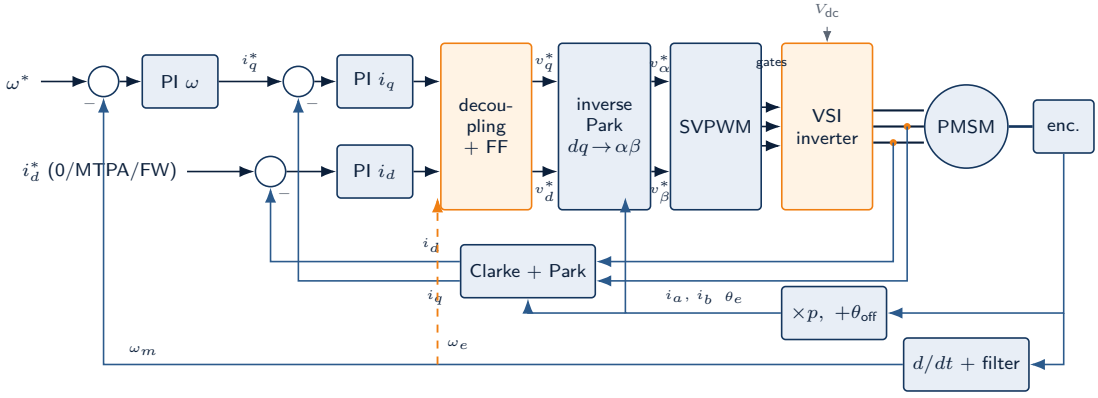
$$P_{\text{el}} = \tfrac{3}{2}(v_d i_d + v_q i_q) = 2.34\,\text{kW}$$

Check: $P_{\text{el}} - P_{\text{mech}} = \tfrac{3}{2} R\,i_q^2 = 75\,\text{W}$ ✓



Vector diagram to scale (voltages 40 V/cm); $\varphi = 13.6°$, $\cos\varphi = 0.97$. The $\omega_e L i_q$ term grows with load and speed – it consumes voltage reserve.

FOC: the basic structure



Every PWM period: sample i_a, i_b and θ_m → Clarke/Park → two PIs → decoupling → inverse Park → SVPWM duty update.

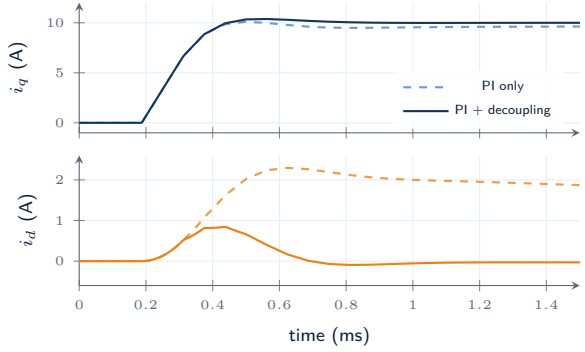
Velocity loop: every 1–4 PWM periods. The outer position loop (Session 3.4) provides ω^*. Typical current-loop execution: 5–20 µs on a DSP/MCU.

Cross-coupling and decoupling

- Without compensation the q-loop must *integrate up* the back-EMF and both PIs fight the coupling terms $\omega_e L\, i$.
- **Decoupling / feedforward**: add the model terms to the PI outputs:

$$v_d^* = v_{d,\mathrm{PI}} - \omega_e \hat{L}_q\, i_q$$
$$v_q^* = v_{q,\mathrm{PI}} + \omega_e(\hat{L}_d\, i_d + \hat{\psi}_{\mathrm{PM}})$$

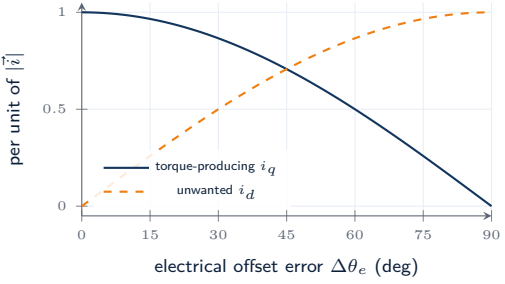
- Ideal decoupling leaves two independent $1/(R + Ls)$ plants.
- Parameter errors and the loop delay leave a residual coupling (right: i_d peak 2.3 A $\rightarrow$ 0.85 A).
- Use i^* instead of i to avoid feeding noise.



Simulated i_q^* step $0 \rightarrow 10$ A at 3000 rpm (reference motor, discrete PI, 16 kHz, 1.5 T_s delay; integrator pre-loaded with $\omega_e \psi_{\mathrm{PM}}$ in both cases).

Rotor angle alignment and offset errors

- The encoder zero is arbitrary; FOC needs $\theta_e = 0$ on the magnet d-axis: $\theta_e = p(\theta_m - \theta_{m,0})$.
- **Methods**
  - DC alignment: inject a current vector along α, rotor locks to $\theta_e = 0$, store $\theta_{m,0}$ (needs free shaft, light load).
  - Back-EMF: spin the motor externally, compare phase-voltage zero crossings with encoder.
  - Hall sensors for start-up, refined at the first Hall edge.
  - Absolute encoder with factory-stored offset (typical for servo motors).
  - Standstill methods with high-frequency injection (saliency / saturation).



$T_e = k_t|\vec{i}|\cos\Delta\theta_e$. $\Delta\theta_m = 2°$ mech. with $p = 4$ $\Rightarrow 8°$ el.: $-1\,\%$ torque, but $14\,\%$ of the current in d (heating, FW in one direction only). For $|\Delta\theta_e| > 90°$ torque reverses $\Rightarrow$ **runaway**.

Summary 3.3 – Field-oriented control

Key take-aways

- Clarke + Park map AC phase quantities into DC dq quantities in the rotor frame; $\theta_e = p\,\theta_m$.
- PMSM dq model: two R–L circuits + cross-coupling $\omega_e L\, i$ + back-EMF $\omega_e \psi_{\text{PM}}$.
- $T_e = \frac{3}{2} p[\psi_{\text{PM}} i_q + (L_d - L_q) i_d i_q]$; SPM: $i_d^* = 0$, IPM: MTPA with $i_d < 0$.
- Field weakening trades current for voltage above base speed.
- FOC = angle + Park + 2 PI + decoupling + inverse Park + SVPWM, every PWM period.
- Angle offset errors cost torque and can cause runaway.

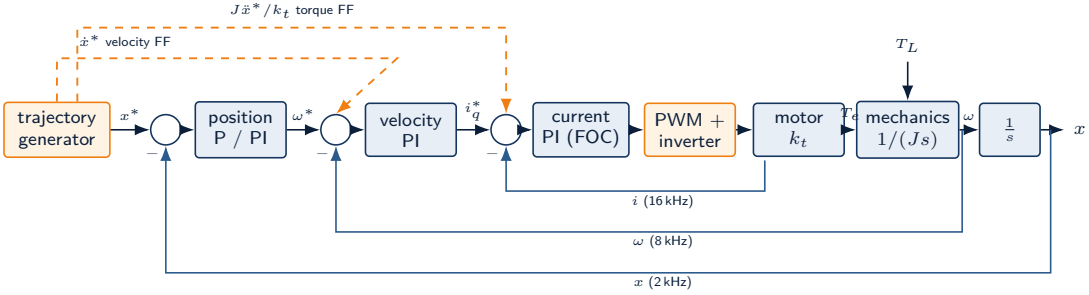
Self-check

1. $i_a = 8$ A, $i_b = -4$ A: i_α, i_β? $(8;\ 0)$
2. Why does i_q step cleanly while i_a is sinusoidal?
3. Back-EMF of the reference motor at 1500 rpm? (75.4 V peak phase)

Session 3

3.4 Current, Velocity and Position Loops

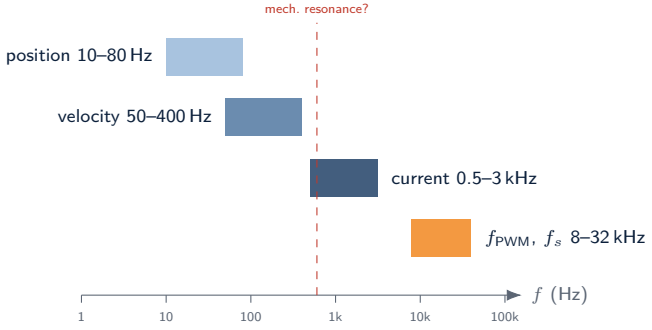
The nested (cascade) control structure



- Each loop sees the closed inner loop as its "actuator".
- Inner loops are faster, sampled faster, and **limit** the variables of the outer loop (current → torque, velocity).
- Tuned and commissioned *inside out*.
- Feedforward paths (dashed) carry the known reference derivatives past the slow loops.

[Ellis, 2012]; [Lorenz, Lipo & Novotny, 1994]

Interaction between loops: the bandwidth hierarchy



- Outer loop bandwidth must stay well below the inner one (factor 3–10) so the inner loop looks like a **unity gain** at the outer crossover.
- Otherwise inner-loop phase lag erodes the outer phase margin.
- The current loop is bounded by sampling and PWM delays; the velocity loop by current loop, sensor filtering and **mechanical resonances**.
- The position loop is bounded by the velocity loop and by permissible overshoot.

Reference and feedback signals

Loop	Reference	Feedback	Sensor / processing	Typical rate
Current	i_d^*, i_q^* (A)	i_d, i_q	shunts / Hall / isolated $\Sigma\Delta$, sampled at PWM centre, Clarke+Park	8–32 kHz
Velocity	ω^* (rad/s)	$\hat{\omega}$	encoder position differences ($\Delta\theta/T_s$), low-pass or observer	4–16 kHz
Position	x^* (rad, mm)	x	motor encoder (semi-closed) or linear scale (full-closed)	1–8 kHz
Feedforward	$\dot{x}^*$, $\ddot{x}^*$	–	from trajectory generator (exact derivatives)	as position

Velocity quantisation

Encoder $N = 2^{20}$ counts/rev, $T_s = 125\,\mu s$: one count $\Rightarrow \Delta\omega = \frac{2\pi}{NT_s} = 0.048$ rad/s. With a 2 500-line (10 000 counts) encoder: 5.0 rad/s – velocity noise forces a filter, the filter adds delay.

Full-closed loop (linear scale on the load) removes screw errors but puts the compliance *inside* the loop $\Rightarrow$ lower achievable bandwidth. Common compromise: velocity from motor encoder, position from scale (dual-loop).

PI current-loop design: pole–zero cancellation

Plant (one axis, after decoupling) and PI controller:

$$G_i(s) = \frac{1}{R + Ls}\, e^{-sT_\sigma}, \qquad C_i(s) = K_p\, \frac{1 + sT_i}{sT_i}$$

Choose $T_i = L/R$ (zero cancels the electrical pole):

$$L_i(s) = \frac{K_p}{Ls}\, e^{-sT_\sigma} \;\Rightarrow\; \omega_c \approx \frac{K_p}{L}$$

- $T_\sigma \approx 1.5\, T_s$: one sample computation delay + ZOH ($T_s/2$); add filter delays.
- Technical optimum (TO): $K_p = \dfrac{L}{2T_\sigma} \Rightarrow \zeta = 0.707$, overshoot $4.3\,\%$, PM $\approx 65^\circ$.

Worked example (reference motor)

$f_s = 16\,\text{kHz}$, $T_s = 62.5\,\mu\text{s}$, $T_\sigma = 93.75\,\mu\text{s}$

$T_i = L/R = 3\,\text{mH}/0.5\,\Omega = 6\,\text{ms}$

$K_p = \dfrac{3\,\text{mH}}{2 \cdot 93.75\,\mu\text{s}} = 16\,\text{V/A}, \quad K_i = K_p/T_i = 2667\,\text{V/(A s)}$

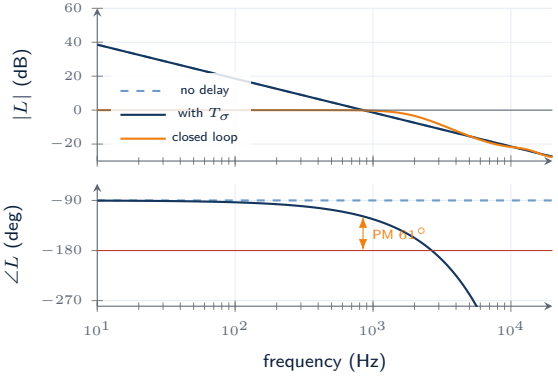
$\omega_c \approx K_p/L = 5333\,\text{rad/s} \rightarrow f_c \approx 850\,\text{Hz}$

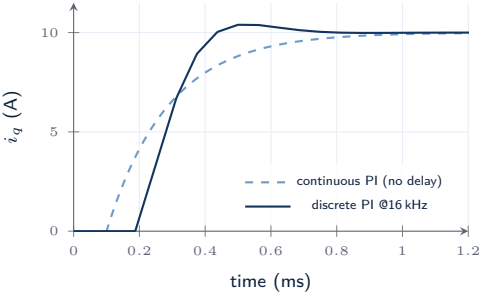
Phase cost of delay at f_c: $360^\circ f_c T_\sigma = 28.7^\circ \Rightarrow$ PM $\approx 61^\circ$

Closed-loop $-3\,\text{dB}$ bandwidth $\approx 1.9\,\text{kHz}$

Discrete implementation: $K_i T_s$ per sample; scale by the inverter gain (V per duty count) and by $\hat{L}$ uncertainty (saturation lowers L at high current!).

Current loop: computed frequency and step response





Step $i_q^* : 0 \rightarrow 10\,\text{A}$ at 0.1 ms. Without delay: first-order, $\tau = L/K_p = 0.19\,\text{ms}$. With sampling and $1.5\,T_s$ delay: overshoot $3.9\,\%$ – close to the TO prediction of $4.3\,\%$.

Velocity PI design: symmetric optimum

Plant seen by the velocity controller:

$$G_\omega(s) = \frac{k_t}{Js} \cdot \frac{1}{1 + sT_{\text{eq}}}, \quad T_{\text{eq}} = 2T_\sigma + T_f$$

($2T_\sigma$: closed current loop; T_f: velocity filter). PI: $C_\omega = K_{p\omega}\frac{1+sT_{i\omega}}{sT_{i\omega}}$.

Symmetric optimum (parameter $a > 1$):

$$\omega_c = \frac{1}{aT_{\text{eq}}}, \quad K_{p\omega} = \frac{J}{a\,k_t\,T_{\text{eq}}}, \quad T_{i\omega} = a^2 T_{\text{eq}}$$

PM $= \arctan\frac{a^2-1}{2a}$: $a = 2 \to 37°$, $a = 3 \to 53°$

Worked example

$T_{\text{eq}} = 2 \cdot 93.75\,\mu\text{s} + 0.31\,\text{ms} = 0.5\,\text{ms}$
$J = J_m(1+3) = 1.2 \times 10^{-3}\,\text{kg m}^2$, $k_t = 0.72\,\text{Nm/A}$

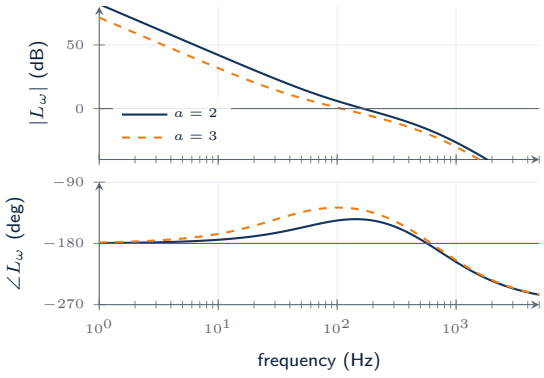
$a = 2$: $K_{p\omega} = \dfrac{1.2 \cdot 10^{-3}}{2 \cdot 0.72 \cdot 5 \cdot 10^{-4}} = 1.67\,\text{A s/rad}$,
$T_{i\omega} = 2\,\text{ms}$, $\omega_c = 1000\,\text{rad/s}$ (159 Hz)

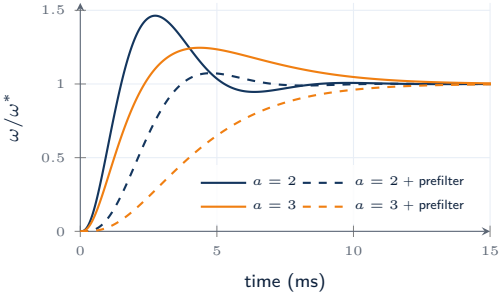
$a = 3$: $K_{p\omega} = 1.11\,\text{A s/rad}$, $T_{i\omega} = 4.5\,\text{ms}$, $f_c = 106\,\text{Hz}$

- The PI zero lies *below* ω_c by factor a, the lag $1/(1 + sT_{\text{eq}})$ above by $a \Rightarrow$ phase is symmetric around ω_c.
- Large overshoot to steps (zero!) $\Rightarrow$ reference prefilter $1/(1 + sT_{i\omega})$.

[Kessler, 1958]; [Schröder, 2015]

Velocity loop: computed Bode plot and step responses





Computed: $a = 2$: $f_c = 164$ Hz, PM 35°, overshoot 46 % (7 % with prefilter). $a = 3$: $f_c = 107$ Hz, PM 53°, 25 % (0 % with prefilter). Small-signal only – no current limit.

Position loop: P (or PI) and the k_v factor

- With the closed velocity loop ≈ 1, the position plant is an integrator $1/s \Rightarrow$ a **P controller** already gives zero steady-state error to steps (type-1 loop).
- $K_{pp} = k_v$ (1/s) is the position-loop crossover; rule: $k_v \approx \omega_{c\omega}/4 \ldots /10$.
- Ramp ($\dot{x}^* = v$ const.) $\Rightarrow$ constant **following error**

$$e_{ss} = \frac{v}{k_v} \qquad \text{(P position loop, no feedforward)}$$

- PI position control removes the ramp error but adds overshoot; usually feedforward is preferred.
- No overshoot allowed in machining (would cut into the part) $\Rightarrow$ P + FF is the standard.

Worked example

Machine tool axis, $k_v = 3\,(\text{m/min})/\text{mm} = 50\,\text{s}^{-1}$, feed $v = 10\,\text{m/min}$:

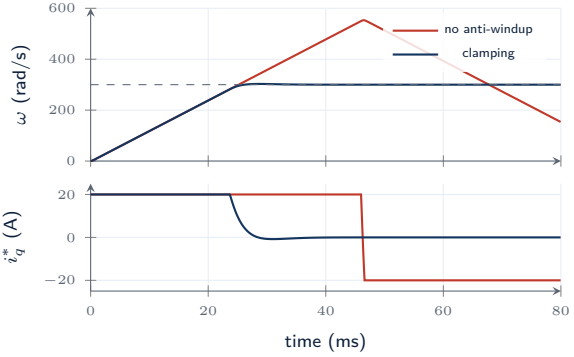
$$e_{ss} = \frac{10}{3} = 3.3\,\text{mm} \text{ (!)}$$

Reference servo, $k_v = 100\,\text{s}^{-1}$, $\omega = 300\,\text{rad/s}$:
$e_{ss} = 3\,\text{rad} \approx$ half a revolution.

On a contour (two axes), different k_v per axis $\Rightarrow$ path distortion. Interpolating axes must have **identical** k_v, or the error must be cancelled by feedforward.

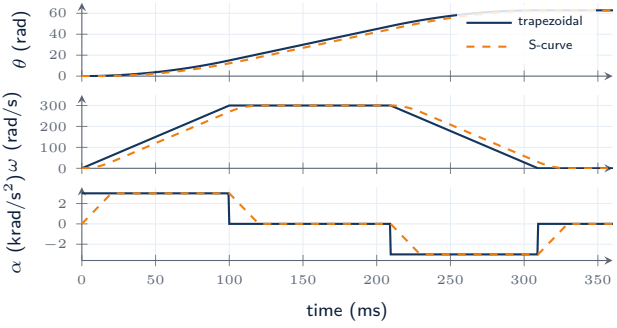
Torque and velocity limits; integrator windup

- Limits: $|i_q^*| \le \hat{I}_{\max}$ (torque), $|\omega^*| \le \omega_{\max}$, and the voltage limit of the current loop.
- While saturated the loop is **open**; a free-running integrator keeps integrating the error $\Rightarrow$ large overshoot (windup).
- Anti-windup: **clamping** (stop integrating when saturated and error drives further) or **back-calculation** ($\dot{I} \mathrel{+}= K_{aw}(u_{\text{sat}} - u)$).
- Applies to all PIs: current (voltage limit), velocity (current limit), position.



Simulated 300 rad/s step, reference motor, $\hat{I}_{\max} = 20$ A, SO $a = 3$: overshoot 553 rad/s vs. 303 rad/s.

Trajectory generation: trapezoidal vs. S-curve profiles

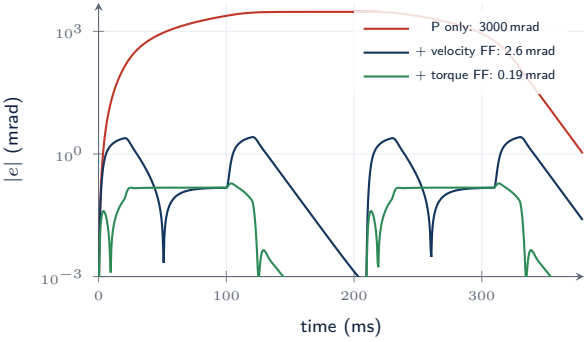


Computed: move 10 rev (20π rad), $\omega_{\max} = 300$ rad/s, $\alpha_{\max} = 3000$ rad/s^2, S-curve jerk $j = 1.5 \times 10^5$ rad/s^3.

- **Trapezoidal**: $t_a = \omega_{\max}/\alpha_{\max} = 0.1$ s, total 309 ms. Acceleration jumps $\Rightarrow$ infinite jerk excites resonances.
- **S-curve** (7 segments): jerk-limited, $t_j = \alpha/j = 20$ ms, total 329 ms (+6 %) but much smoother.
- Peak torque $J\alpha_{\max} = 3.6$ Nm ($i_q = 5$ A) – within limits.
- Triangular profile if the distance is too short to reach $\omega_{\max}$.
- Generator outputs $x^*, \dot{x}^*, \ddot{x}^*$ at the position-loop rate.

[Biagiotti & Melchiorri, 2008]

Following error and how feedforward removes it



Simulated cascade (current loop, SO velocity PI $a = 2$, $k_v = 100\,\mathrm{s}^{-1}$) on the S-curve move; legend: peak $|e|$. Log scale!



5-axis machining centre. G. McKechnie, CC BY-SA 2.0, via Wikimedia Commons

- Velocity FF $\omega^* \mathrel{+}= \dot{x}^*$: loop no longer has to create the speed from error.
- Torque/acceleration FF $i_q^* \mathrel{+}= \hat{J}\ddot{x}^*/k_t$: inertia no longer needs velocity error.
- FF does *not* change stability – it acts outside the loop; accuracy depends on $\hat{J}$ and on smooth (S-curve) references.

Summary 3.4 – Cascade loops

Key take-aways

- Cascade: position (P) → velocity (PI) → current (PI), bandwidths separated.
- Current PI: $T_i = L/R$, $K_p = L/(2T_\sigma)$ (TO) ⇒ ≈ 850 Hz crossover at 16 kHz.
- Velocity PI: symmetric optimum $\omega_c = 1/(aT_{\text{eq}})$, prefilter for steps.
- P position loop: following error v/k_v.
- Limits require anti-windup; S-curves limit jerk.
- Velocity + torque feedforward reduce following error by orders of magnitude.

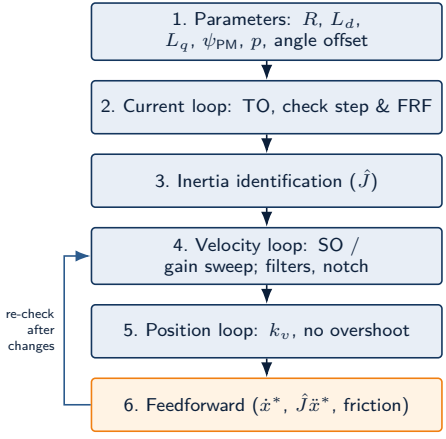
Self-check

1. $L = 6$ mH, $T_\sigma = 100$ µs: TO gain? (30 V/A)
2. Following error at $v = 0.2$ m/s, $k_v = 40\,\text{s}^{-1}$? (5 mm)
3. Why is a prefilter needed with SO tuning?

Session 4

3.5 Loop Tuning and Bandwidth

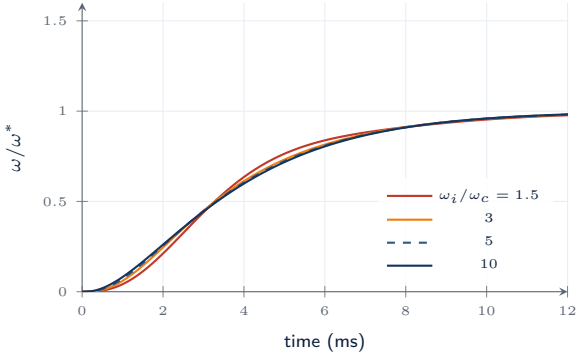
Tuning procedure: from the inside out



- Each outer loop is designed on the **closed** inner loop – tune outer before inner and you tune against the wrong plant.
- Measure, do not guess: auto-tuning routines identify R, L (voltage steps), $\hat{J}$ (acceleration test: $\hat{J} = k_t\,\Delta i_q / \Delta\dot{\omega}$) and FRFs.
- Increase gains until a margin limit is hit (PM $\geq 45°$, GM $\geq 6\,\text{dB}$, $|S|_{\max} \leq 2$, i.e. 6 dB), then back off.
- Check large-signal behaviour last (limits, anti-windup, saturation of L).
- Record final FRFs as a commissioning baseline.

[Ellis, 2012]; [Åström & Hägglund, 2006]

Bandwidth separation: rule of thumb 3–10×



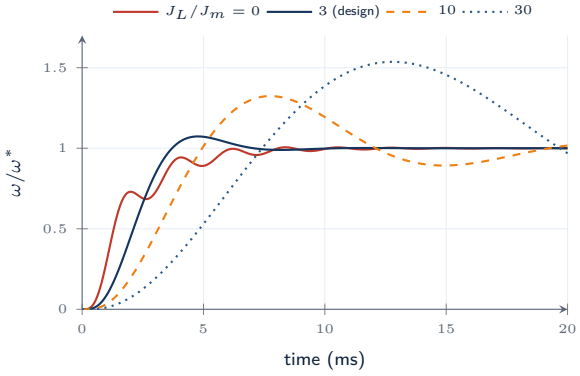
Computed: velocity PI designed for $\omega_c = 1000$ rad/s *assuming an ideal current loop* ($K_p = J\omega_c/k_t$, $T_i = 4/\omega_c$, prefilter); the real inner loop is $1/(1 + s/\omega_i)$.

- A first-order inner loop with bandwidth ω_i costs $\arctan(\omega_c/\omega_i)$ phase at the outer crossover:

ω_i/ω_c	phase lost
1.5	$33.7°$
3	$18.4°$
5	$11.3°$
10	$5.7°$

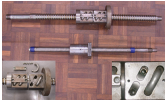
- Below 3×: outer design must include the inner dynamics explicitly (as SO does via T_{eq}).
- Above 10×: wasteful – the outer loop could be faster.

Effect of the load inertia ratio



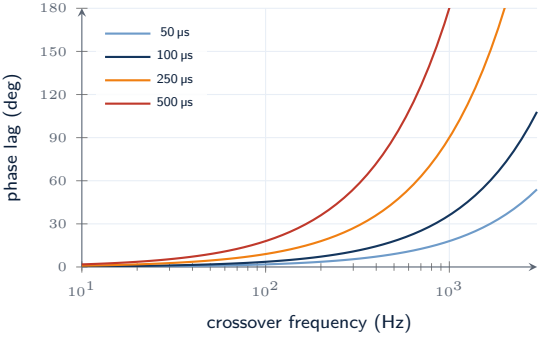
Gains fixed (SO, $a = 2$, designed for ratio 3). Computed open-loop crossover / PM: 0: 439 Hz/12°; 3: 164 Hz/35°; 10: 80 Hz/31°; 30: 43 Hz/21°.

- Loop gain $\propto K_{p\omega} k_t / J$: the same gains give a **lower crossover** with more load inertia and a higher one (towards the delays!) with less.
- Rule: retune with $\hat{J}$; adaptive gain scaling $K_{p\omega} \propto \hat{J}$.
- Typical recommendation: $J_L/J_m \leq 3 \ldots 10$ for high dynamics – otherwise gearbox or larger motor.



Ball screws. Graibeard, CC BY-SA 3.0, via Wikimedia Commons

Delays and their phase cost at crossover



Pure delay e^{-sT_d}: $\Delta\varphi = 360^\circ\, f_c\, T_d$ – linear in f, no magnitude change.

Delay source	typical
Computation (1 sample at 16 kHz)	62.5 µs
PWM / ZOH ($T_s/2$)	31 µs
Current sensing, $\Sigma\Delta$ decimation	5–20 µs
Encoder serial protocol	10–30 µs
Velocity filter (T_f)	0.1–1 ms
Field bus (set-point, 1 cycle)	0.125–1 ms

Example

Velocity loop, $T_{\text{eq}} = 0.5$ ms at $f_c = 160$ Hz: 28.8° lost. Halving T_f or doubling f_s buys phase that can be traded for bandwidth.

Technical optimum vs. symmetric optimum

	Technical (magnitude) optimum, TO	**Symmetric optimum, SO**
Plant type	$\dfrac{K}{(1+sT_1)(1+sT_\sigma)}$, $T_1 \gg T_\sigma$	$\dfrac{K}{sT_I\,(1+sT_\sigma)}$ (integrating)
PI choice	$T_i = T_1$, $K_p = \dfrac{T_1}{2KT_\sigma}$	$T_i = a^2 T_\sigma$, $K_p = \dfrac{T_I}{aKT_\sigma}$
Closed loop	$\approx \dfrac{1}{1+2T_\sigma s+2T_\sigma^2 s^2}$, $\zeta = 0.707$	zero at $1/T_i$, needs prefilter
Step response	4.3 % overshoot, $t_r \approx 4.7\,T_\sigma$	43 % ($a=2$), 8 % with prefilter
Phase margin	$\approx 65^\circ$	$\arctan\frac{a^2-1}{2a}$ (37° at $a=2$)
Disturbance rejection	slow for $T_1 \gg T_\sigma$ (pole cancelled)	good: no cancellation
Used for	**current loop** ($T_1 = L/R$)	**velocity loop** (inertia = integrator)

Both are *starting points* from German drive practice (Kessler, 1955/58). Industrial auto-tuners do similar things in the frequency domain with explicit PM/GM targets.

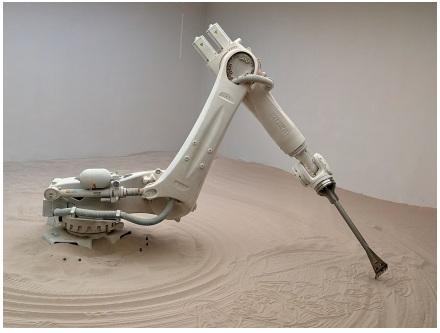
For the velocity loop many practitioners use $a = 2.5 \ldots 4$ (PM 45–60°) for robustness to J and resonances.

Typical bandwidths of industrial servo drives

Loop	closed-loop BW	limited by
Current	1–3 kHz	f_s, PWM delay
Velocity	100–500 Hz (rigid) 20–100 Hz (compliant)	filters, resonance
Position	10–100 Hz	velocity loop, overshoot
Machine-tool k_v	1–6 (m/min)/mm	17–100 s^{-1}

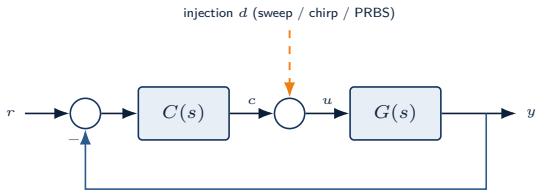
Order-of-magnitude values from drive manuals and the literature [Ellis, 2012]; the achievable values depend on the mechanics far more than on the drive.

- Robots: velocity BW limited by gear compliance (strain wave gears) to a few tens of Hz per joint.
- Direct drives (linear/torque motors): velocity BW several hundred Hz.



Industrial robot. Oleg Yunakov, CC BY-SA 4.0, via Wikimedia Commons

Measuring frequency response functions (FRFs)



$$G = \frac{Y}{U}, \qquad S = \frac{U}{D}, \qquad L = -\frac{C}{U} = \frac{1}{S} - 1$$

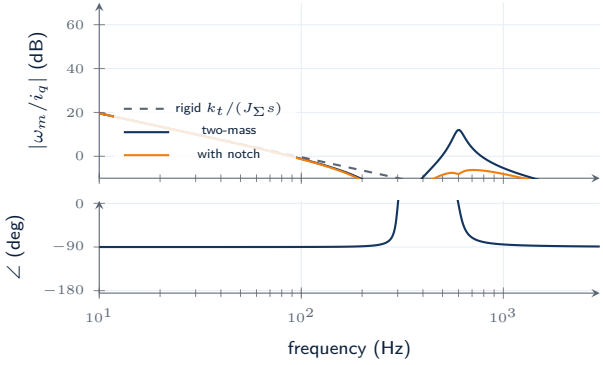
- Excite *inside* the closed loop (machine stays under control); estimate with cross-spectra $\hat{G} = \hat{S}_{uy}/\hat{S}_{uu}$, average, check **coherence** $\gamma^2 \approx 1$.
- Injection at current reference $\Rightarrow$ mechanical plant (k_t/Js + resonances); at velocity reference $\Rightarrow$ closed-loop BW.
- Amplitude: large enough above friction/noise, small enough to stay linear (no current limit).



HP 3562A dynamic signal analyzer. Loxyger, CC BY 4.0, via Wikimedia Commons

Modern drives have this analyzer built in: the commissioning software injects a chirp and plots Bode diagrams directly.

A measured-style FRF: two-mass resonance and notch filter



Computed: $J_m = 3 \times 10^{-4}$, $J_L = 9 \times 10^{-4}$ kg m^2, $k_s = 3200$ Nm/rad, 3 % damping.

- Anti-resonance $f_a = \frac{1}{2\pi}\sqrt{k_s/J_L} = 300$ Hz, resonance $f_r = f_a\sqrt{1 + J_L/J_m} = 600$ Hz.
- Above f_r the motor "sees" only J_m: gain jumps by $1 + J_L/J_m$ (12 dB) $\Rightarrow$ crossover may move up into the delays $\Rightarrow$ instability ("buzz").
- **Notch filter** at f_r (in the current reference): $N(s) = \dfrac{s^2 + 2\zeta_z\omega_r s + \omega_r^2}{s^2 + 2\zeta_p\omega_r s + \omega_r^2}$, here $\zeta_z = 0.05$, $\zeta_p = 0.5$.
- Notch costs phase below f_r; resonances move with load $\Rightarrow$ deeper treatment on Day 5.

Summary 3.5 – Tuning and bandwidth

Key take-aways

- Tune inside out; each loop on the closed inner loop.
- Separate bandwidths 3–10×; less needs explicit modelling.
- The inertia ratio scales the velocity-loop gain: retune with $\hat{J}$.
- Every delay costs $360^\circ f_c T_d$ – budget them.
- TO for the current loop, SO for the velocity loop as starting points.
- Measure FRFs in closed loop; watch coherence, resonances, notches.

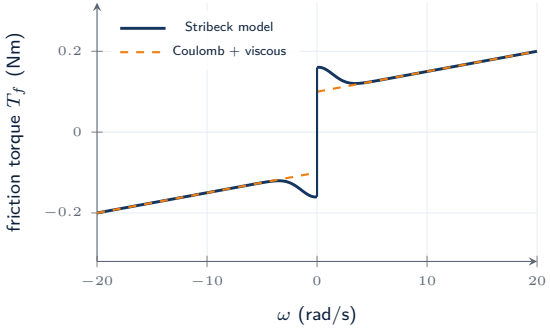
Self-check

1. Phase lost by 200 µs at 250 Hz? (18°)
2. Inertia ratio doubles: what happens to ω_c with fixed gains?
3. Why inject at the controller output rather than open the loop?

Session 5

3.6 Disturbances and Compensation

Disturbances: load torque and friction

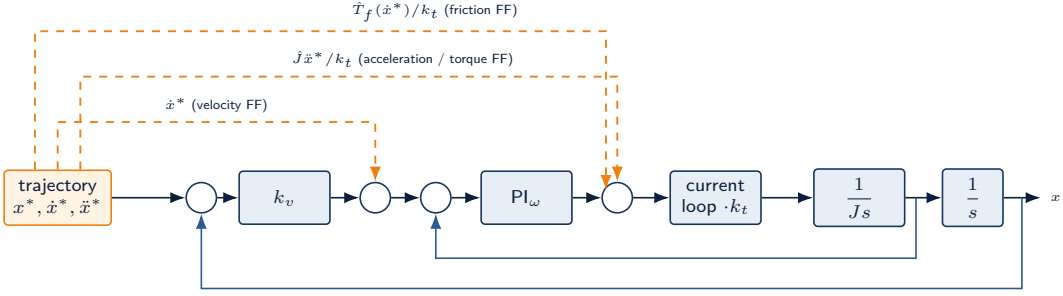


$T_f = \mathrm{sign}(\omega)\left[T_C + (T_S - T_C)e^{-(\omega/\omega_S)^2}\right] + B\omega$; here $T_C = 0.1$ Nm, $T_S = 0.16$ Nm, $\omega_S = 2$ rad/s, $B = 5$ mNm s/rad.

- Mechanics: $J\dot{\omega} = k_t i_q - T_L - T_f(\omega)$.
- **Load torque** T_L: process forces, gravity (vertical axes!), cutting forces.
- **Friction**: discontinuous at $\omega = 0 \Rightarrow$ stick-slip, quadrant glitches on circles, limit cycles with integral action.
- PI rejects a step T_L with zero steady-state error, but the transient dip depends on ω_c and J.
- Other disturbances: cogging and ripple torque, dead-time voltage error, measurement noise, cable forces.

[Armstrong-Hélouvry, Dupont & Canudas de Wit, 1994]

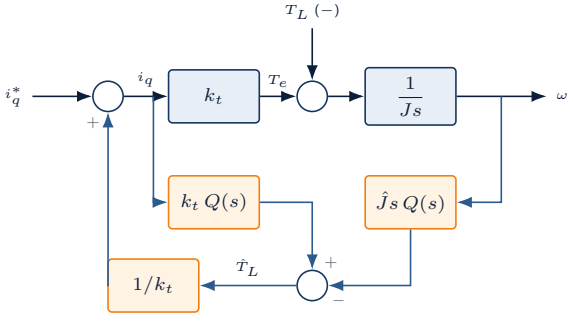
Feedforward: velocity, acceleration/torque and friction



- Ideal FF inverts the plant: $u_{\mathrm{FF}} = G^{-1}x^*$. For a rigid axis: $i_q^* = \left(\hat{J}\ddot{x}^* + \hat{B}\dot{x}^* + \hat{T}_f\right)/k_t$.
- Requires smooth references (S-curve) – FF of a jerk step is a torque step.

- FF acts **outside the loop**: no stability change, feedback only corrects the model error.
- Typical gains 80–100 % of model; check on overshoot at the end of acceleration.

Disturbance observer (DOB): basic logic



$$Q(s) = \frac{1}{1 + s\tau_Q}$$ low-pass makes the inverse model $\hat{J}s$ proper.

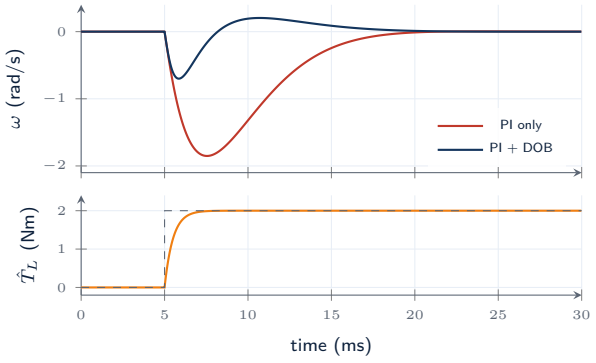
- Idea: compare what the motor *should* do ($k_t i_q$) with what it *does* ($\hat{J}\dot{\omega}$). The difference is the lumped disturbance:

$$\hat{T}_L = Q(s)\left[k_t i_q - \hat{J}s\,\omega\right]$$

- Feeding $\hat{T}_L/k_t$ back makes the axis behave like the nominal $1/(\hat{J}s)$ below $1/\tau_Q$ – load, friction and inertia error are rejected.
- Bandwidth of Q limited by noise ($s\omega$!) and delays; typical $1/\tau_Q = 0.5$–$2\times\omega_c$.
- $\hat{T}_L$ is also a useful signal: load monitoring, collision detection.

[Ohnishi, Shibata & Murakami, 1996]

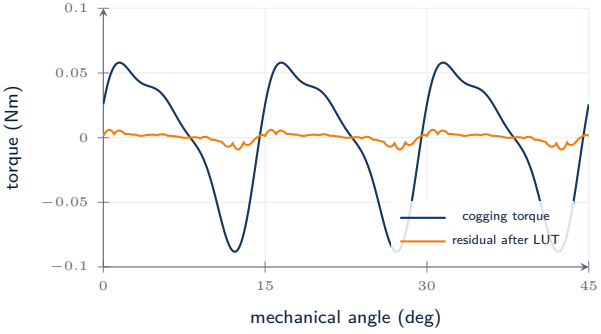
DOB vs. PI alone: load-torque step



Simulated: reference motor, velocity PI (SO, $a = 3$) holding $\omega^* = 0$, load step $T_L = 2\,\text{Nm}$ at 5 ms; DOB with $\tau_Q = 0.5\,\text{ms}$, $\hat{J} = J$.

- Peak speed dip: 1.85 rad/s → 0.70 rad/s (2.6× better), and recovery is faster.
- The estimate $\hat{T}_L$ converges to the true 2 Nm within $\approx 3\tau_Q$ plus loop delays.
- Same effect could be obtained with more PI gain – but that would cost phase margin and noise; DOB adds rejection **without** changing the reference response (nominal).

Cogging and torque ripple compensation (brief)



Illustrative 12-slot/8-pole motor: fundamental cogging period $360^\circ/\mathrm{LCM}(12,8) = 15^\circ$ mech. LUT with 1° steps, 95 % gain: 0.146 Nm p-p $\rightarrow$ 0.015 Nm p-p.

- **Cogging**: magnets vs. slot openings, present at zero current, periodic in θ_m.
- **Ripple**: back-EMF harmonics, current sensor offsets/gain errors (1st/2nd order in θ_e), dead time (6th).
- Compensation: measure i_q needed at constant slow speed vs. angle $\Rightarrow$ look-up table $\Delta i_q(\theta_m)$ added to i_q^*; or iterative learning / repetitive control.
- Effective at low speed, where the velocity loop cannot reject the ripple frequency $pn/60\times$ harmonic order.
- Sensor offset calibration first – it is the cheapest ripple reduction.

Summary 3.6 – Disturbances and compensation

Key take-aways

- Load torque and friction act at the plant input; PI rejects them with limited speed.
- Feedforward (velocity, acceleration/torque, friction) cancels *known* effects outside the loop.
- DOB estimates the *unknown* lumped disturbance from $k_t i_q - \hat{J}\dot{\omega}$ and compensates it.
- Cogging/ripple: periodic in angle $\Rightarrow$ table-based compensation.

Self-check

1. Which compensation for a vertical axis with gravity load? (constant torque FF / integrator / DOB)
2. Why does DOB need $Q(s)$?
3. Cogging period of a 9-slot/6-pole motor? (20°)

Session 6

Day 4 Wrap-up

Day summary

From voltage to motion

3.2 SVPWM synthesises $\vec{v}^*$ up to $V_{\mathrm{dc}}/\sqrt{3}$; dead time and f_{sw} shape the voltage quality; current & voltage limits bound the dq plane.

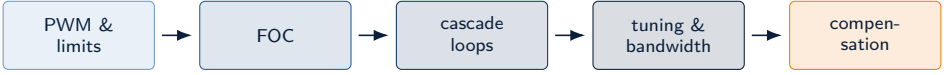
3.3 FOC makes the PMSM a DC-like actuator: $T_e = \frac{3}{2}p[\psi_{\mathrm{PM}} i_q + (L_d - L_q) i_d i_q]$, MTPA, field weakening, decoupling, correct angle.

From loops to accuracy

3.4 Cascade P–PI–PI, TO/SO design, limits with anti-windup, S-curve trajectories.

3.5 Inside-out tuning, 3–10× separation, inertia ratio, delay budget, FRF measurement.

3.6 Feedforward and disturbance observers remove what feedback alone cannot.



Preview of Day 5

Resonance, Sensors, Measurement and System Integration

- 3.7 Resonance, vibration and the use of filters – continuing today's two-mass FRF and notch.
- 4.1–4.2 Sensors; encoders and resolvers – where θ_m and $\hat{\omega}$ really come from.
- 4.3–4.4 Sampling, digitisation and signal processing – quantisation, aliasing, filter delay.
- 4.5 Communication and system integration – field buses, synchronisation of set-points.

Bring along from today
– the delay budget table,
– the velocity-loop plant $k_t/(Js)$,
– the encoder quantisation example,
– the two-mass FRF.

Figure credits

All plots, block diagrams, vector diagrams and schematics without a credit line are original drawings computed for this course (data generated by `day4/src/gen_data.py`). Downloaded figures (Wikimedia Commons):

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